## Type MS · Linear inequalities and the 'both' overlap

Answer 1 of 15 · Question 1 part a · 2 marks

2023 JAN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

**Method marks for differentiation and integration:**

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

**Use of a formula:**

Generally, the method mark is gained by **either**

Type MS · Linear inequalities and the 'both' overlap

Answer 1 of 15 · Question 1 part a · 2 marks

2023 JAN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

**or**, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

**Answers without working:**

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."**Exact answers:**

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

**Rounding answers (where accuracy is specified in the question)**

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

## Type MS · Linear inequalities and the 'both' overlap

Answer 1 of 15 · Question 1 part a · 2 marks

2023 JAN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

| **Paper 2** | | |
|---|---|---|
| Question number | Scheme | Marks |
| 1 a | $9x < 6 \Rightarrow x < \ldots$ | M1 |
| | $x < \frac{2}{3}$ | A1<br>(2) |
| b | $(3x+1)(x-3) < 0$ | M1 |
| | $x = -\frac{1}{3} \quad x = 3$ | A1 |
| | $-\frac{1}{3} < x < 3$ | M1 A1<br>(4) |
| c | $-\frac{1}{3} < x < \frac{2}{3}$ | B1ft<br>(1) |
| | | **Total 7 marks** |

Type MS · Linear inequalities and the 'both' overlap

Answer 2 of 15 · Question 1 part a · 1 mark

2022 JAN · Paper 02 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

**Method marks for differentiation and integration:**

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Type MS · Linear inequalities and the 'both' overlap

Answer 2 of 15 · Question 1 part a · 1 mark

2022 JAN · Paper 02 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

**Use of a formula:**

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

**or**, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

**Answers without working:**

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Type MS · Linear inequalities and the 'both' overlap

Answer 2 of 15 · Question 1 part a · 1 mark

2022 JAN · Paper 02 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

**Exact answers:**

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

**Rounding answers (where accuracy is specified in the question)**

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

## Type MS · Linear inequalities and the 'both' overlap

### Answer 2 of 15 · Question 1 part a · 1 mark

2022 JAN · Paper 02 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 1 (a) | $x > 3$ | B1<br>(1) |
| (b) | $(2x-1)(x-5)$ | M1 |
| | Critical values are $x = \frac{1}{2}$ and $x = 5$ | A1 |
| | $\frac{1}{2} < x < 5$ | B1ft<br>(3) |
| (c) | $3 < x < 5$ | B1ft<br>(1) |
| | | **Total 5 marks** |

Type MS · Linear inequalities and the 'both' overlap

Answer 3 of 15 · Question 2 part a · 1 mark

2023 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

| | | |
|---|---|---|
| **2(a)** | $3x-8<5x+3 \Rightarrow x>-\frac{11}{2}$ | B1<br>[1] |
| **(b)** | $4x^2-7x+1>6-2x^2 \Rightarrow$<br>$6x^2-7x-5>0 \Rightarrow (3x-5)(2x+1)>0$<br>cv's are $x=\frac{5}{3},\ -\frac{1}{2} \Rightarrow x<-\frac{1}{2},\ x>\frac{5}{3}$ | M1M1<br><br>M1A1<br>[4] |
| **(c)** | $-\frac{11}{2}<x<-\frac{1}{2},\ x>\frac{5}{3}$ | B1 ft<br>[1] |
| | **Total** | **6 marks** |

| Question | Notes | Marks |
|---|---|---|
| **2(a)** | $3x-8<5x+3$<br>$\Rightarrow x>-\frac{11}{2}$ | B1<br>[1] |
| **(b)** | For rearranging to form a 3TQ<br>$4x^2-7x+1>6-2x^2 \Rightarrow 6x^2-7x-5>0$ | M1 |
| | For method to solve their 3TQ. See general guidance for what constitutes an attempt to solve.<br>$6x^2-7x-5>0 \Rightarrow (3x-5)(2x+1)>0$ | M1 |
| | For writing down their critical values from their factorisation and the correct inequality for their cv's [outside region].<br>cv's are $x=\frac{5}{3},\ -\frac{1}{2}$<br>$x<-\frac{1}{2},\ x>\frac{5}{3}$ | M1 |
| | For the correct inequalities only $x<-\frac{1}{2},\ x>\frac{5}{3}$<br>A0 if rejects one region or if inequalities are incorrectly combined. | A1<br>[4] |
| **(c)** | For the correct inequality only $-\frac{11}{2}<x<-\frac{1}{2},\ x>\frac{5}{3}$<br>For follow through (a) must be a linear inequality and (b) must be of the form $x<a, x>b$ with $b>a$. | B1 ft<br>[1] |
| | **Total** | **6 marks** |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 3 of 15 · Question 2 part a · 1 mark

2023 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

### Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 1 of 23 · find part a in this mark scheme extract

| | | |
|---|---|---|
| 2(a) | $x < \frac{1}{2}$ o.e | B1<br>(1) |
| (b) | $10x^2 + 7x - 12 (< 0)$<br>$(5x - 4)(2x + 3) (< 0)$<br><br>$x = \frac{4}{5}$ $\quad$ $x = -\frac{3}{2}$<br><br>$-\frac{3}{2} < x < \frac{4}{5}$ o.e | M1<br><br><br><br>M1A1<br>(3) |
| (c) | $-\frac{3}{2} < x < \frac{1}{2}$ | B1FT<br>(1) |
| | | **Total 5 marks** |

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | B1 | $x < \frac{1}{2}$ o.e |
| (b) | M1 | Rearranges to a 3TQ and attempts to solve. See general guidance for what constitutes an attempt to solve.<br>If they use a calculator to solve the 3TQ, the 3TQ must be correct and the values of $x$ also correct.<br>They must obtain **two** values of $x$ for this mark |
| | M1 | For forming an **inside** region with their critical values. |
| | A1 | For the correct inequality.<br>May be seen described in set notation. |
| (c) | B1FT | For $-\frac{3}{2} < x < \frac{1}{2}$<br>FT their (a) and (b) provided they overlap and it is an inside region.<br><br>**NB** The final answer given must be consistent with their parts (a) and (b) |

| Question | Scheme | Marks |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

### Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 2 of 23 · find part a in this mark scheme extract

| number | | |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 3 of 23 · find part a in this mark scheme extract

| | |
|---|---|
| M1 | Multiplies numerator and denominator by $6+2\sqrt{5}$ with no errors |
| M1 | Equates numerator coefficients to find two simultaneous equations in $a$ and $b$ |
| M1 | Solves their simultaneous equations by any method provided the equations of the form;<br>$ma+nb=...$ and $pa+qb=...$ where $m \neq p$ and $n \neq q$<br>They must reach both a value for $a$ and a value for $b$ for this mark |
| M1 | For multiplying through by 8 or 4 to give a prime number in the denominator of $\frac{9+4\sqrt{5}}{c}$ |
| A1 | For the correct values of $a$, $b$ and $c$<br>Allow embedded values. |
| **ALT** | |
| M1 | 'Cross multiplies' through by $\left(6-2\sqrt{5}\right)$ **AND** by $c$ |
| M1 | Multiplies out both sides with no errors |
| M1 | Collects up like terms to obtain $c\left(a+b\sqrt{5}\right)=A+B\sqrt{5}$ |
| M1 | Factorises their expression to obtain $P\left(Q+R\sqrt{5}\right)$ where $P$, $Q$ and $R$ are integers<br>or deduces by inspection that $a = 7$ and $c = 2$ |
| A1 | For the correct values – accept if embedded in the expression. |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 4 of 23 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 4 (a) (i) | $\frac{ar^6}{ar^2}=\frac{40}{10}$ | M1 |
| | $r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$ | M1 |
| | $r=-\sqrt{2}$ | A1 |
| (ii) | $ar^2=10\Rightarrow a=\frac{10}{'2'}$ | M1 |
| | $a=5$ | A1 (5) |
| (b) | $\lvert r\rvert>1$ so the series is not convergent. | B1FT (1) |
| | | **Total 6 marks** |

| Part | Mark | Additional Guidance |
|---|---|---|
| (a)(i) | M1 | For $\frac{ar^6}{ar^2}=\frac{40}{10}$ or $\frac{ar^2}{ar^6}=\frac{40}{10}$ |
| | M1 | For rearranging to find a value for $r$<br>$r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$ or $\left(\pm\sqrt[4]{4}\right)$ |
| | A1 | For the correct value of $r=-\sqrt{2}$ Accept $-\sqrt[4]{4}$ |
| (a)(ii) | M1 | For attempting to find a value for $a$ using their $r$<br>$ar^2=10\Rightarrow a=\frac{10}{'2'}$ |
| | A1 | For the correct value of $a=5$<br>Allow this mark from the use of $r=\sqrt{2}$ |
| (b) | B1FT | $\lvert r\rvert>1$ so the series is not convergent.<br>Follow through their value of $r$ provided it is greater than 1. |

Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 5 of 23 · find part a in this mark scheme extract

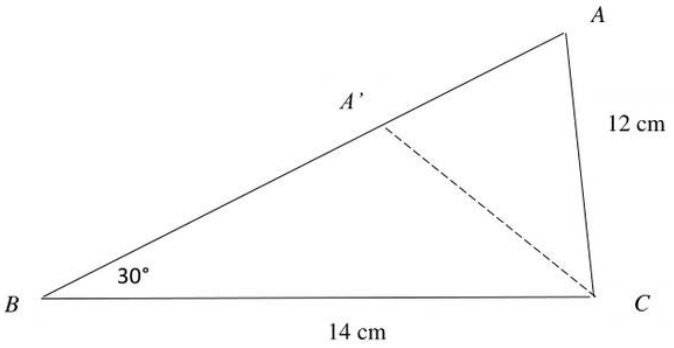
| Question number | Scheme | Marks |
|---|---|---|
| 5(a) | $12^2 = 14^2 + x^2 - 2\times 14x\cos 30$ | M1 |
| | $x^2 - 14\sqrt{3}x + 52 = 0$ | M1 A1 |
| | $x = \dfrac{14\sqrt{3} \pm \sqrt{(14\sqrt{3})^2 - 4(1)(52)}}{2(1)}$ | M1 |
| | $x = 7\sqrt{3} \pm \sqrt{95}$ | A1cso [5] |
| (b) | Area $= \dfrac{1}{2}\times 14\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \sin 30$ or uses Area $= \dfrac{1}{2}\times 12\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \dfrac{7}{12}$ | M1 |
| | Difference in areas | |
| | $D = \dfrac{1}{2}\times 14\times\left(7\sqrt{3} + \sqrt{95}\right)\times \sin 30 - \dfrac{1}{2}\times 14\times\left(7\sqrt{3} - \sqrt{95}\right)\times \sin 30$ | dM1 |
| | $D = 7\sqrt{95}$ | A1 [3] |
| **ALT Part (a) – sine rule** | | |
| | $\dfrac{12}{\sin 30} = \dfrac{14}{\sin A} \Rightarrow \sin A = \dfrac{7}{12}$ | {M1 |
| | $\left(\dfrac{7}{12}\right)^2 + \cos^2 A \equiv 1 \Rightarrow \cos A = \sqrt{1 - \left(\dfrac{7}{12}\right)^2}$ | M1 |
| | $\sin A = \dfrac{7}{12}, \cos A = \pm\dfrac{\sqrt{95}}{12}$ | A1 |
| | $x = 12\cos A + 14\cos 30 \Rightarrow x = 12\times\dfrac{\pm\sqrt{95}}{12} + 14\times\dfrac{\sqrt{3}}{2}$ | M1 |
| | $x = 7\sqrt{3} \pm \sqrt{95}$ * | A1 Cso [5]} |
| | | **Total 8 marks** |

Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 6 of 23 · find part a in this mark scheme extract

**USEFUL SKETCH**




| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | Use the cosine rule to form an equation. Rule to be correct but may be in any rearranged form. |
| | M1 | Rearrange to obtain a 3TQ. Terms in any order. = 0 may be missing. |
| | A1 | Correct 3TQ Allow poor notation – missing = 0 |
| | M1 | Attempt to solve their 3TQ by use of the formula or completing the square. See general guidance for what constitutes an attempt to solve. Use of calculators: If the 3TQ is correct, accept correct solutions without working. |
| | A1 | Correct values for $x$ $x=7\sqrt{3}\pm\sqrt{95}$ |
| (b) | M1 | For use of area of triangle $=\frac{1}{2}ac\sin B$ with $a=14$, $B=30°$ and *their* $x$ May use either value found for $x$. Or uses $=\frac{1}{2}bc\sin A$ with $c=12$ and $\sin A=\frac{7}{12}$ |
| | dM1 | For finding the difference between their two areas provided both areas are positive This M mark is dependent on the previous M mark |
| | A1 | Correct difference in areas (must be exact value) |

| | ALT Part (a) – Uses sine rule | |
|---|---|---|
| | M1 | Use of the sine rule to find sin$A$ |
| | M1 | Use of $\sin^2 A+\cos^2 A\equiv 1$ to find an exact value for $\cos A$ |
| | A1 | Correct exact values for sin$A$ and $\cos A$ |
| | M1 | Method to find $x$ $x=12\cos A+14\cos 30$ Allow *their* $\cos A$ |
| | A1 | Correct values for $x$ $x=7\sqrt{3}\pm\sqrt{95}$ |

## Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 7 of 23 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 6 (a) | $700\pi = 2\pi r^2 + 2\pi rh$ | B1 |
| | $h = \dfrac{350 - r^2}{r}$ or $rh = 350 - r^2$ | M1 |
| | $V = \pi r^2 h = \pi r^2\left(\dfrac{350 - r^2}{r}\right) = \left\{\pi r\left(350 - r^2\right)\right\} = \pi r\left(350 - r^2\right)$* | M1A1* cso [4] |
| (b) | $V = 350\pi r - \pi r^3$ | |
| | $\dfrac{dV}{dr} = 350\pi - 3\pi r^2 = 0$ | M1 |
| | $\Rightarrow 350\pi = 3\pi r^2 \Rightarrow r = \sqrt{\dfrac{350}{3}} = 10.8$ (3sf) | M1A1 |
| | $\dfrac{d^2V}{dr^2} = -6\pi r$ | M1 |
| | $\dfrac{d^2V}{dr^2} < 0 \therefore$ max | A1 [5] |
| (c) | $h = \dfrac{350 - r^2}{r} = \dfrac{350 - \frac{350}{3}}{\sqrt{\frac{350}{3}}} = 21.6$ (3 s.f.) | B1 [1] |
| | | **Total 10 marks** |

## Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 8 of 23 · find part a in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | B1 | For a **correct** equation for the surface area in terms of the radius and height |
| | M1 | For rearranging to find $h$ in terms of $r$ which must be as a minimum<br>$h = \frac{k - lr^2}{mr}$ where $k$, $l$ and $m$ are integers<br>OR<br>$rh = k - lr^2$ where $k$ and $l$ are integers |
| | M1 | For a correct expression for volume using *their* $h$ in terms of $r$ |
| | A1* cso | For rearranging and obtaining the given result with no errors or omissions and must include $V = \ldots$<br>[Allow Volume = …. or Vol = …..] |
| (b) | M1 | For an attempt to differentiate the **given** expression for $V$ which must be of the form $\frac{dV}{dr} = 350\pi - K\pi r^2$ where $K$ is a constant<br><br>Using Product Rule:<br><br>$\frac{dV}{dr} = \pi r(-2r) + \pi(350 - r^2) \Rightarrow \frac{dV}{dr} = \left[350\pi - K\pi r^2\right]$ |
| | M1 | For setting their $\frac{dV}{dr} = 0$ **and** attempting to find a value for $r$. |
| | A1 | $r = \sqrt{\frac{350}{3}}$ [=10.801...] accept awrt 10.8 |
| | M1 | For finding $\frac{d^2V}{dr^2}$<br>Accept only $\frac{d^2V}{dr^2} = \pm k\pi r$ where $k$ is a constant<br>See general guidance for what constitutes an attempt to differentiate. |
| | A1 | For the correct conclusion derived from the correct working throughout. |
| (c) | B1 | For obtaining the value of $h$ correct to 3 s.f.<br>Accept awrt 21.6 |

Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 9 of 23 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 7 (a) | $x$: 0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5<br>$y$: 4.26, **4.14**, 4, **3.83**, 3.63, **3.37**, 3, 2.37 | B2<br>(2) |
| (b) | Points plotted within half a square.<br>Points joined with a smooth curve. | B1ft<br>B1ft<br>(2) |
| (c) | $3^{2x-5} = (4-x)^3 \Rightarrow 2x-5 = 3\log_3(4-x)$<br>$\frac{2x-5}{3}+3 = \log_3(4-x)+3 \Rightarrow \frac{2x+4}{3} = \log_3(4-x)+3$<br>**ALT**<br>$\log_3(4-x)+3 = mx+c$<br>$\Rightarrow \log_3(4-x) = mx+(c-3)$<br>$\Rightarrow (4-x) = 3^{mx+(c-3)}$<br>$\Rightarrow (4-x)^3 = 3^{(mx+(c-3))3} = 3^{3mx+3c-9}$<br>$\Rightarrow 2x-5 = 3mx+3c-9 \Rightarrow 3m = 2$ and $-5 = 3c-9$<br>$\Rightarrow m = \frac{2}{3} \quad c = \frac{4}{3}$<br>Graph of $y = \frac{2x+4}{3}$ drawn. Intersection point is $x = 2.8$ | M1M1<br>M1A1<br><br><br>[M1<br>M1<br>M1<br>A1]<br>M1A1<br>(6)<br>**[10]** |

## Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 10 of 23 · find part a in this mark scheme extract

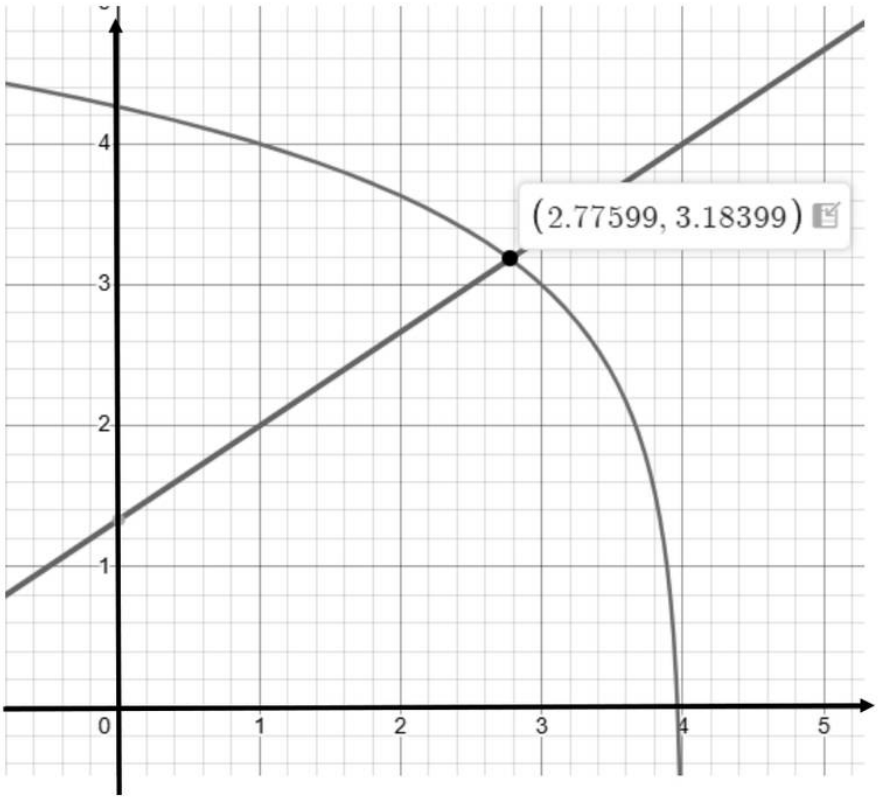
| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | B2 | All three points correct<br>(B1 for 2 points correct) |
| (b) | B1ft | Points plotted ft their table. Allow half a square tolerance. |
| | B1ft | Points joined together with a smooth curve. Allow half a square tolerance on joining up their points.<br>FT their table |
| (c) | M1 | For taking logs base 3 of both sides<br>$2x-5=\log_3(4-x)^3$ |
| | M1 | For applying the power law<br>$2x-5=3\log_3(4-x)$ |
| | ddM1 | For attempting to rearrange to the required form<br>An attempt is defined as dividing through by 3 and adding 3 to both sides. |
| | A1 | For the correct line, in any form<br>$\frac{2x-5}{3}+3=\log_3(4-x)+3\Rightarrow\left[\left(y=\right)\frac{2}{3}x+\frac{4}{3}\right]$ |
| | **ALT for first 4 marks** | |
| | M1 | Sets up the equation to obtain $\log_3(4-x)+3=mx+c$<br>And removes the log to obtain $(4-x)=3^{mx+(c-3)}$ |
| | M1 | Raises **both** sides to a power of 3.<br>Allow one error in this on the LHS |
| | M1 | Compares coefficients to obtain an equation in $m$ and an equation in $c$ |
| | A1 | Obtains the values $m=\frac{2}{3}$ $c=\frac{4}{3}$ |
| | **Draws the line** | |
| | M1 | For their line drawn provided it is of the form $y=kx\pm\frac{4}{3}$ for $k\neq 0,\ 1$ |
| | A1 | $x=2.8$ or 2.7<br>Calculator value is 2.776 |

## Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 11 of 23 · find part a in this mark scheme extract

**USEFUL GRAPH**



Type MS · Linear inequalities and the 'both' overlap

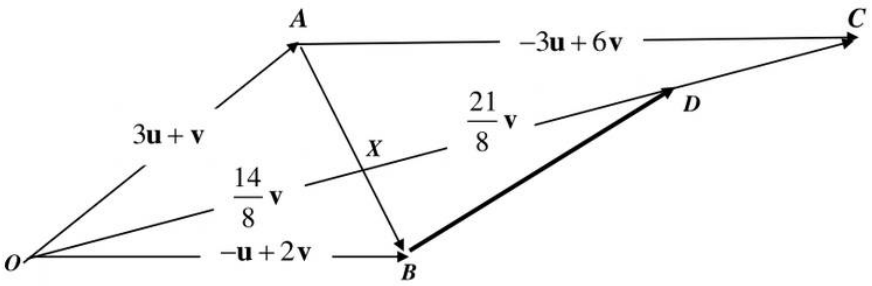
Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 12 of 23 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 8 (a) | $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}, = -(3\mathbf{u}+\mathbf{v}) + (2\mathbf{v}-\mathbf{u}) = -4\mathbf{u}+\mathbf{v}$ | M1,A1<br>(2) |
| (b) | $\overrightarrow{AC} = \lambda(2\mathbf{v}-\mathbf{u})$ | M1 |
| | $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = (3\mathbf{u}+\mathbf{v}) + \lambda(2\mathbf{v}-\mathbf{u}) = \mu\mathbf{v}$ | M1 |
| | $\mathbf{u}: 0 = 3-\lambda \qquad \mathbf{v}: \mu = 1+2\lambda$ | M1 |
| | $\mu = 7 \qquad [\lambda = 3]$ | A1<br>(4) |
| (c) | $\overrightarrow{OX} = \overrightarrow{OA} + \alpha\overrightarrow{AB} = 3\mathbf{u}+\mathbf{v}+\alpha(-4\mathbf{u}+\mathbf{v})$<br>or<br>$\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AC} + \beta\overrightarrow{CO} = 3\mathbf{u}+\mathbf{v}+3(2\mathbf{v}-\mathbf{u})+\beta(-7\mathbf{v})$ | M1 |
| | $\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$ | A1 |
| | $\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$ | M1 |
| | $\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$ | M1 |
| | $\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{35}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{7}{4}\mathbf{v}+\frac{21}{8}\mathbf{v}$<br>$\overrightarrow{BD} = \mathbf{u}+\frac{19}{8}\mathbf{v}$ | A1<br>(5) |
| | | **Total 11 marks** |

**USEFUL SKETCH**

$$\overrightarrow{OD} = \frac{35}{8}\mathbf{v} \quad \text{and} \quad \overrightarrow{AX} = -3\mathbf{u}+\frac{3}{4}\mathbf{v}$$



Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 13 of 23 · find part a in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | Use of $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$ |
| | A1 | $-4\mathbf{u} + \mathbf{v}$ |
| (b) | M1 | For use of $\overrightarrow{AC}$ is parallel to $\overrightarrow{OB}$ to form an expression for $\overrightarrow{AC}$ |
| | M1 | For use of their expression for $\overrightarrow{AC}$ to form an expression for $\overrightarrow{OC}$ |
| | M1 | For equating for **u** and for **v** |
| | A1 | For $\mu = 7$ |
| (c) | | **General principles for marking part (c)**<br>For example: You may see the vector path for $\overrightarrow{BD}$ first - score the 4th M mark<br>• Candidates will need to find vector $\overrightarrow{OX}$ [M1A1]<br>• Candidates will need to use the ratio of bases to find<br>$\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$ **leading** to establish<br>$\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$ [M1]<br>• Check any path they choose on the diagram below. For example,<br>$\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OX} + \overrightarrow{XD}$<br>or [M1]<br>$\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OA} + \overrightarrow{AX} + \overrightarrow{XD}$<br>If a candidate chooses to take a longer path and need to find for example, $\overrightarrow{AX}$ they will need to find a vector. There is no separate mark for this.<br>• For the correct vector |
| | M1 | For an expression for $\overrightarrow{OX}$ including an unknown constant |
| | A1 | For $\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$ |
| | M1 | For recognising that triangle $OBX$ and triangle $OBD$ share a base $OB$ and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{OD}$ together with an attempt to find $\overrightarrow{OD}$<br>OR<br>For recognising that triangle $OBX$ and triangle $BXD$ share a base $BX$ and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{XD}$ together with an attempt to find $\overrightarrow{XD}$ |
| | M1 | For method to find $\overrightarrow{BD}$ Award this for any valid path given |
| | A1 | $\overrightarrow{BD} = \mathbf{u} + \frac{19}{8}\mathbf{v}$ |

Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 14 of 23 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 9 (a) | $\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ * | M1A1A1 Cso [3] |
| (b) | **Method A**<br>$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$<br>$\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$<br>$2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$<br>$\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y \Rightarrow \frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$<br>$\Rightarrow \frac{d^2y}{dt^2} = -8\frac{dy}{dt} - 20y \Rightarrow \frac{d^2y}{dt^2} + 8\frac{dy}{dt} + 20y = 0$<br>$\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = 0 \Rightarrow M = 8, N = 20$ | M1<br>M1<br>M1<br>M1<br>A1 [5] |
| | **Method B**<br>$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$<br>$\left(= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t\right)$<br>$\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$<br>$= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$<br>$= (16 - 2M)e^{-4t}\sin 2t + (12 - 4M + N)e^{-4t}\cos 2t$<br>$16 - 2M = 0$ and $12 - 4M + N = 0$<br>$M = 8,\ N = 20$ | [M1<br>M1<br>M1<br>M1<br>A1] |
| | **Method C**<br>By inspection – please see additional guidance. | |
| | | **Total 8 marks** |

| Part | Mark | Additional Guidance |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

### Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 15 of 23 · find part a in this mark scheme extract

| | | |
|---|---|---|
| (a) | M1 | For an attempt to differentiate the given expression.<br>• There needs to be an acceptable attempt to differentiate both terms. $e^{-4t}\cos 2t \rightarrow ke^{-4t}\sin 2t + le^{-4t}\cos 2t$ with $k, l \neq 0$<br>• There need to be two terms added.<br>$\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t$ |
| | A1 | One term completely correct.<br>$\frac{dy}{dt} = -4e^{-4t}\cos 2t + '-2e^{-4t}\sin 2t$ OR $\frac{dy}{dt} = '-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t$ |
| | A1 | Fully correct differentiated expression and rearranged into the required form<br>$\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$<br>Accept also,<br>For example:<br>$-\left(-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t\right) - 4\left(e^{-4t}\sin 2t\right) = 2e^{-4t}\sin 2t$<br>With any unambiguous indication that a conclusion has been reached, e.g., QED, hash sign, double underline, RHS = LHS etc. |
| | | **Method A** |
| (b) | M1 | For an attempt to find $\frac{d^2y}{dt^2}$<br>Minimally acceptable attempt is<br>$ke^{-4x}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$<br>$k, l$ as in their first derivative, $m, n, p, q \neq 0$<br>$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ |
| | M1 | For substituting $y$ and $\frac{dy}{dt}$ into their $\frac{d^2y}{dt^2}$<br>$\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$ |
| | M1 | For preparing to eliminate $\cos 2t$ by rearranging their $\frac{dy}{dt}$<br>$2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$<br>Allow errors in arithmetic but not mathematically incorrect process. |
| | M1 | For an unsimplified expression only in terms of $y$, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$<br>$\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$ |
| | A1<br>[8] | For finding the value of $M$ and the value of $N$.<br>$M = 8,\ N = 20$ |
| | | **Method B** |

## Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

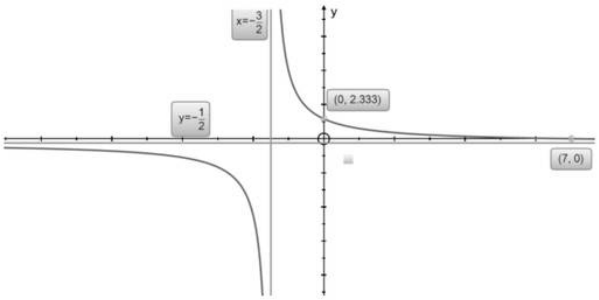
2025 JUN · Paper 02 · extract 16 of 23 · find part a in this mark scheme extract

| | | |
|---|---|---|
| | M1 | For an attempt to find $\frac{d^2y}{dt^2}$<br>Minimally acceptable attempt is<br>$ke^{-4t}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$<br>$k, l$ as in their first derivative, $m, n, p, q \neq 0$<br>$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ |
| | M1 | For substituting $y$, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ into the equation.<br>$\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$<br>$= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$ |
| | M1 | For collecting terms<br>$\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = ('16' - 2M)e^{-4t}\sin 2t + ('12' - 4M + N)e^{-4t}\cos 2t$ |
| | M1 | For identifying two equations from their working and a method to solve them simultaneously<br>For example, $'16' - 2M = 0$ and $'12' - 4M + N = 0$ |
| | A1 | For finding the value of $M$ and the value of $N$.<br>$M = 8, N = 20$ |
| | | **Method C (By inspection)** |
| | M1 | For an attempt to find $\frac{d^2y}{dt^2}$<br>Minimally acceptable attempt is<br>$ke^{-4t}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$<br>$k, l$ as in their first derivative, $m, n, p, q \neq 0$<br>$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ |
| | M1 | For simplifying $\frac{d^2y}{dt^2} = e^{-4t}16\sin 2t + e^{-4t}12\cos 2t = \left[e^{-4t}(16\sin 2t + 12\cos 2t)\right]$ |
| | M1 | For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}[\sin 2t]$ and deducing that $'16' + (\underline{8} \times -2) = 0 \Rightarrow M = 8$ |
| | M1 | For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}$ and $y$ and deducing that $'12' - ('8' \times 4) + \underline{20} = 0 \Rightarrow N = 20$ |
| | A1 | For both $M = 8$ and $N = 20$ |
| | | **Total 8 marks** |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 17 of 23 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 10 (a) | (i) $x=-\frac{3}{2}$ (ii) $y=-\frac{1}{2}$ | B1<br>B1<br>[2] |
| (b) | $(7,0)$<br>$\left(0,\frac{7}{3}\right)$ | B1<br>B1<br>[2] |
| (c) |  | B1ft curve<br>B1ft asymptotes<br>B1ft (intersections with $x$- and $y$-axes)<br>[3] |
| (d) | $\frac{-1(2x+3)-2(7-x)}{(2x+3)^2}=\left[\frac{-17}{(2x+3)^2}\right]$ | M1 |
| | $\frac{-1\left(2\times\frac{1}{2}+3\right)-2\left(7-\times\frac{1}{2}\right)}{\left(2\times\frac{1}{2}+3\right)^2}=-\frac{17}{16}$ | M1A1* cso<br>[3] |
| | ALT – product rule | |
| | $-1(2x+3)^{-1}+(7-x)(-1)(2)(2x+3)^{-2}=\left[\frac{-17}{(2x+3)^2}\right]$ | M1 |
| | $-1\left(2\times\frac{1}{2}+3\right)^{-1}+\left(7-\frac{1}{2}\right)(-1)(2)\left(2\times\frac{1}{2}+3\right)^{-2}=-\frac{17}{16}$ | M1A1* cso |

## Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 18 of 23 · find part a in this mark scheme extract

| | | |
|---|---|---|
| (e) | $-\frac{17}{(2x+3)^2}=-\frac{17}{16}$ | M1 |
| | $(2x+3)^2=16 \Rightarrow 2x+3=\pm\sqrt{16} \Rightarrow x=0.5 \text{ or } -3.5$ | dM1 |
| | $y=\frac{7--3.5}{2\times-3.5+3}\left(=-\frac{21}{8}\right)$ | ddM1 |
| | $\left(-\frac{7}{2},-\frac{21}{8}\right)$ | A1 |
| | $y--\frac{21}{8}=-\frac{17}{16}\left(x--\frac{7}{2}\right)$ | M1A1 |
| | $34x+32y+203=0$ | A1<br>[7] |
| | | **Total 17 marks** |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 19 of 23 · find part a in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| **For parts (a) and (b) look for their responses in the body of the question. In part (a), if the responses are not labelled (i) and (ii) accept them in order.** | | |
| (a) | (i) B1 | For $x=-\frac{3}{2}$ oe |
| | (ii) B1 | For $y=-\frac{1}{2}$ oe |
| (b) | B1 B1 | First B1 for either correct, second B1 for both correct<br>Condone if not given as coordinates e.g. $x=7$ and/or $y=\frac{7}{3}$ given |
| (c) | B1ft | Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes. |
| | B1ft | Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes. |
| | B1ft | Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around.<br>If additional intersections seen then B0 |
| (d) | M1 | Attempt the quotient rule. Numerator must be of the form $A.(2x+3)-B.(7-x)$, $B>1$.<br>Denominator must be of the form $(2x+3)^2$ |
| | M1 | Substitutes $x=\frac{1}{2}$ into their derivative<br>**NB: This is an A mark in Epen** |
| | A1* cso | Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors<br>**NB: This is a B mark in Epen** |
| | ALT – product rule | |
| | M1 | For an attempt at Product Rule.<br>Must be a sum of two products.<br>Must have the form<br>$c(2x+3)^{-1}+d(7-x)(2x+3)^{-2}$ for constants $c$, $d$. |
| | M1 | Substitutes $x=\frac{1}{2}$ into their derivative<br>**NB: This is an A mark in Epen** |
| | A1* cso | Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors<br>**NB: This is a B mark in Epen** |
| (e) | M1 | For equating the gradient of the tangent to their $\frac{dy}{dx}$ from (d) |
| | dM1 | For rearranging to obtain $(2x+3)^2=16$ and attempt to solve<br>or for rearrangement to a 3TQ and method to solve the 3TQ. See general guidance for acceptable methods to solve. |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 20 of 23 · find part a in this mark scheme extract

| | | |
|---|---|---|
| | ddM1 | For substitution of their $x$ into the original equation to obtain $y$.<br>This must **not** be $x=\frac{1}{2}$<br>This M mark is dependent on the **two** previous M marks |
| | A1 | For correct coordinate of $P\left(-\frac{7}{2},-\frac{21}{8}\right)$ |
| | M1 | For substitution of their coordinates of $P$ and the gradient $-\frac{17}{16}$ to obtain an equation of a straight line using any valid method.<br>If $y=mx+c$ is used a value of $c$ must be found **and** an equation needs to be formed.<br>For example; $'-\frac{21}{8}'=-\frac{17}{16}\times'-\frac{7}{2}'+c\Rightarrow c=-\frac{203}{32}\Rightarrow y=-\frac{17}{16}x-\frac{203}{32}$ [ |
| | A1 | For any correct equation of the tangent with the correct values of $x$ and $y$ in any form. |
| | A1 | For the correct equation of tangent given in the required form.<br>$34x+32y+203=0$ |

Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 21 of 23 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 11 (a) | $\cos^4\theta-\sin^4\theta=(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$ | M1 |
| | $=(\cos^2\theta-\sin^2\theta)\times1=\cos2\theta$ * | M1M1A1 cso [4 |
| | **ALT** example<br>$\cos^4\theta-\sin^4\theta=\cos^4\theta-\sin^2\theta\sin^2\theta$ | |
| | $=\cos^4\theta-(1-\cos^2\theta)(1-\cos^2\theta)$ | 2nd M1 |
| | $=\cos^4\theta-(1-2\cos^2\theta+\cos^4\theta)$ | 1st M1 |
| | $=(1-2\cos^2\theta)$ | 3rd M1 |
| | $=\cos2\theta$ | A1 |
| (b) | $8\cos^2\left(2\theta+\frac{\pi}{4}\right)-3=2[\cos^4\left(\theta+\frac{\pi}{8}\right)-\sin^4\left(\theta+\frac{\pi}{8}\right)]$ | M1 |
| | $=2\cos\left(2\theta+\frac{\pi}{4}\right)$ | A1 |
| | $8\cos^2\left(2\theta+\frac{\pi}{4}\right)-2\cos\left(2\theta+\frac{\pi}{4}\right)-3=0$ | M1 |
| | $\left(4\cos\left(2\theta+\frac{\pi}{4}\right)-3\right)\left(2\cos\left(2\theta+\frac{\pi}{4}\right)+1\right)[=0]$ | |
| | $\cos\left(2\theta+\frac{\pi}{4}\right)=\frac{3}{4}\quad\cos\left(2\theta+\frac{\pi}{4}\right)=-\frac{1}{2}$ | A1 |
| | $2\theta+\frac{\pi}{4}=0.72273...\quad2\theta+\frac{\pi}{4}=\frac{2\pi}{3}$ | dM1 |
| | $\theta=2.39,3.11,\ 0.65,1.70$ accept $\theta=\frac{5\pi}{24},\frac{13\pi}{24}$ | A1A1 [7] |
| (c) | $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos^4 2x-\sin^4 2x-8\sin 4x\right)\mathrm{d}x=\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos 4x-8\sin 4x\right)\mathrm{d}x$ | M1 |
| | $=\left[\frac{1}{4}\sin 4x+\frac{8}{4}\cos 4x\right]_{\frac{\pi}{16}}^{\frac{\pi}{8}}$ | M1 |
| | $=\frac{1}{4}\left[\sin\left(4\times\frac{\pi}{8}\right)-\sin\left(4\times\frac{\pi}{16}\right)\right]+2\left[\cos\left(4\times\frac{\pi}{8}\right)-\cos\left(4\times\frac{\pi}{16}\right)\right]$ | M1 |
| | $=\frac{1}{4}\left[1-\frac{\sqrt{2}}{2}\right]+2\left[0-\frac{\sqrt{2}}{2}\right]=\frac{1}{4}-\frac{9\sqrt{2}}{8}$ | A1 [4] |
| | | **Total 15 marks** |

## Type MS · Linear inequalities and the 'both' overlap

Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 22 of 23 · find part a in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| **General Principles for marking part (a)**<br>• First M mark is for either applying the difference of two squares or multiplying out their expression and collecting terms.<br>• Second M mark is for applying $\cos^2\theta+\sin^2\theta=1$ in any form. This can be implied from correct work. However, do not award for just $\cos^4\theta-\sin^4\theta=\cos^2\theta-\sin^2\theta$ seen.<br>• Third M mark is for applying any identity for $\cos 2\theta$<br>• A1 For obtaining the required result with no errors. | | |
| (a) | M1 | For factorising as the difference of two squares $(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$ |
| | M1 | For use of $\cos^2\theta+\sin^2\theta)=1$ |
| | M1 | For use of $\cos^2\theta-\sin^2\theta=\cos(2\theta)$ |
| | A1 cso | Obtain the given result with no errors in the working. |
| (b) | M1 | For using the given result to obtain a quadratic in $\left(2\theta+\frac{\pi}{4}\right)$<br>Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark |
| | A1 | For rearranging to the correct quadratic |
| | M1 | For a valid attempt to solve their **3TQ**.<br>See General Guidance for what constitutes an attempt to factorise.<br>If candidates use a calculator [so no working shown] then award this mark only if the 3TQ is correct with the correct solutions. |
| | A1 | For obtaining correct values for $\cos\left(2\theta+\frac{\pi}{4}\right)$<br>Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark |
| | dM1 | For achieving any valid value **in radians** for their $2\theta+\frac{\pi}{4}$<br>i.e. 0.7227…or $\frac{2}{3}\pi$<br>Accept values out of range for this mark.<br>This is dependent on the previous M mark. |
| | A1 | For two correct solutions in range |
| | A1 | For all four correct solutions in range and no others.<br>Extra solutions within range scores A0<br>Ignore solutions outside of range.<br>Exact solutions when given must be given as multiples of π. |
| (c) | M1 | For use of the given result to obtain $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\cos 4x\ \mathrm{d}x$ |
| | M1 | For an attempt to integrate $\cos kx\to l\sin kx$ where $k,l\neq 1$ |
| | M1 | For substituting limits into a changed expression |
| | A1 | For the correct answer in the required form. |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 4 of 15 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 23 of 23 · find part a in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

## Type MS · Linear inequalities and the 'both' overlap

### Answer 5 of 15 · Question 1 part a · 2 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

| Number | | |
|---|---|---|
| 1 (a) | $3x < 12$ | M1 |
| | $x < 4$ | A1 |
| | | [2] |
| (b) | $(2x+1)(x-3) > 0$ | M1 |
| | Critical values are $x = -\frac{1}{2}$ and $x = 3$ | M1 |
| | $x < -\frac{1}{2}$ $x > 3$ | A1 |
| | | [3] |
| (c) | $x < -\frac{1}{2}$ $3 < x < 4$ | B1ft |
| | | [1] |
| | | **Total 6 marks** |

## Type MS · Linear inequalities and the 'both' overlap

Answer 5 of 15 · Question 1 part a · 2 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

| Part | Mark | Guidance |
|---|---|---|
| (a) | M1 | Attempts to solve the inequality to achieve $3x < 12$<br>Allow $3x < a$ where $a$ is an integer |
| | A1 | For $x < 4$ |
| (b) | M1 | Attempts to solve the inequality by any method to find critical values<br>See General Guidance for acceptable methods.<br>If a calculator is used, the critical values must be fully correct for this mark.<br><br>Allow = or > for this mark or even no sign at all provided it is clear they are solving a quadratic.<br>$(2x+1)(x-3) > 0 \Rightarrow x = ...,\ ...$ $\left(x = -\frac{1}{2},\ 3\right)$ |
| | M1 | For forming a correct inequality, which must be an open interval, following through their **two** critical values which must have come from the solution of a 3TQ.<br>$x < -\frac{1}{2} \quad x > 3$<br>Accept any correct notation. E.g., $x < -\frac{1}{2}$ or $x > 3$<br>Or $\left\{x : x < -\frac{1}{2}\right\} \cup \{x : x > 3\}$<br>Condone $x < -\frac{1}{2}$ and $x > 3$ for this mark only |
| | A1 | For the correct inequality with the correct critical values using any acceptable notation.<br>Eg, $x < -\frac{1}{2} \quad x > 3$ |
| (c) | B1ft | For $x < -\frac{1}{2} \quad 3 < x < 4$<br>ft their answers from parts (a) and (b)<br>provided (b) is of the form $x < p$ and/or $x > q$<br><br>**Note:** If you have already penalised them for writing $x < -\frac{1}{2}$ **and** $x > 3$ in part (b) then allow $x < -\frac{1}{2}$ **and** $3 < x < 4$ for this mark. |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 5 of 15 · Question 1 part a · 2 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Linear inequalities and the 'both' overlap

Answer 6 of 15 · Question 2 part a · 2 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

| | | |
|---|---|---|
| **2(a)** | $2(x+1)<5x-2 \Rightarrow 2x+2<5x-2 \Rightarrow 3x>4 \Rightarrow x>\frac{4}{3}$ | M1A1<br>[2] |
| **(b)** | $3x^2-x-10=0 \Rightarrow (3x+5)(x-2)=0 \Rightarrow x=-\frac{5}{3}, 2$<br>$'-\frac{5}{3}' \leqslant x \leqslant '2'$ allow $'-\frac{5}{3}' < x < '2'$<br>$-\frac{5}{3} \leqslant x \leqslant 2$ | M1<br>dM1<br>A1<br>[3] |
| **(c)** | $"\frac{4}{3}" < x '\leqslant' "2"$ | B1ft<br>[1] |
| | | **Total 6 marks** |

| Part | Mark | Notes |
|---|---|---|
| (a) | M1 | Attempts to solve the inequality with no more than one arithmetical error. |
| | A1 | For the correct inequality. |
| (b) | M1 | For attempting to solve the QE to find **two** critical values. |
| | For the dM and A marks allow any acceptable notation.<br>For example; $-\frac{5}{3} \leqslant x \cap x \leqslant 2$ or, $-\frac{5}{3} \leqslant x$ **and** $x \leqslant 2$<br>The region must however indicate an INSIDE region | |
| | dM1 | Forms an inside region with **their** cv's<br>This mark is dependent on the first M mark in (b) |
| | A1 | For the correct region with the correct inequalities. |
| (c) | B1ft | For the correct combined inside region.<br>Penalise the incorrect inequality from part (b) $<$ in place of $\leqslant$ only once.<br>Even if the **correct** inequality does not follow from their work, award this mark. |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 6 of 15 · Question 2 part a · 2 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

### Answer 7 of 15 · Question 2 part a · 1 mark

2020 JAN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 2(a) | $x \leqslant -1$ | B1 | (1) |
| (b) | $8x^2 + 10x - 3 (< 0)$ | | |
| | $(4x - 1)(2x + 3) (< 0)$ | M1 | |
| | $x = \frac{1}{4} \quad x = -\frac{3}{2}$ | A1A1 | |
| | $-\frac{3}{2} < x < \frac{1}{4}$ | A1ft | (4) |
| (c) | $-\frac{3}{2} < x \leqslant -1$ | B1 | (1) |
| | | | **[6]** |

| | |
|---|---|
| (a) | |
| B1 | For $x \leqslant -1$ |
| (b) | Accept decimals in (b) and (c) |
| **NB** | The first 3 marks are for finding the critical values. Allow with < or = used. |
| M1 | Attempt to obtain the critical values by solving their 3TQ by any valid method. |
| A1 | Either CV correct |
| A1 | Second CV correct. Award these 2 marks if correct CVs seen in an inequality. |
| A1ft | Inequality formed to indicate the values between their CVs. Must use < (Can be written in set language). |
| **NB** | If CVs incorrect and only shown in the inequality, award 0/4 if no working shown for solving their 3TQ: if working shown M1A0A0A1 is available. |
| (c) | |
| B1 | For $-\frac{3}{2} < x \leqslant -1$ (no ft) |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 7 of 15 · Question 2 part a · 1 mark

2020 JAN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

| Question Number | Scheme | Marks |
|---|---|---|

Type MS · Linear inequalities and the 'both' overlap

Answer 8 of 15 · Question 1 part whole · 4 marks

2024 JUN · Paper 01 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

1. Factorisation:

$(x^2 + bx + c) = (x + p)(x + q)$,where $|pq| = |c|$ leading to $x =$….

$(ax^2 + bx + c) = (mx + p)(nx + q)$ where$|pq| = |c|$ and $|mn| = |a|$leading to $x =$….

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x =$….

3. Completing the square:

$x^2 + bx + c = 0$: $\left(x \pm \frac{b}{2}\right)^2 \pm q \pm c = 0, \quad q \neq 0$ leading to $x =$….

**Method marks for differentiation and integration:**

1. Differentiation

Power of at least one term decreased by 1. $(x^n \rightarrow x^{n-1})$

2. Integration:

Power of at least one term increased by 1. $(x^n \rightarrow x^{n+1})$

**Use of a formula:**
Generally, the method mark is gained by **either** quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values **or**, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

**Answers without working:**
The rubric states "Without sufficient working, correct answers may be awarded no marks". General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

**Exact answers**
When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

**Rounding answers (where accuracy is specified in the question)**
Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the is rule may allow the mark to be awarded before the final answer is given.

## Type MS · Linear inequalities and the 'both' overlap

Answer 8 of 15 · Question 1 part whole · 4 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|
| 1 | $\sqrt{50}x-\sqrt{18}>6x+5\Rightarrow\sqrt{50}x-6x>5+\sqrt{18}\Rightarrow x>\dfrac{5+\sqrt{18}}{\sqrt{50}-6}$ | M1 |
| | $\dfrac{5+\sqrt{18}}{\sqrt{50}-6}\times\dfrac{\sqrt{50}+6}{\sqrt{50}+6}=\dfrac{5\sqrt{50}+30+\sqrt{18}\sqrt{50}+6\sqrt{18}}{14}=\dfrac{43\sqrt{2}+60}{14}$ | M1dM1 |
| | $x>\dfrac{43\sqrt{2}+60}{14}$ | A1<br>[4] |
| | | **Total 4 marks** |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 8 of 15 · Question 1 part whole · 4 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

Answer 9 of 15 · Question 1 part c · 1 mark

2022 JAN · Paper 02 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

**Method marks for differentiation and integration:**

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Type MS · Linear inequalities and the 'both' overlap

Answer 9 of 15 · Question 1 part c · 1 mark

2022 JAN · Paper 02 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

**Use of a formula:**

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

**or**, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

**Answers without working:**

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Type MS · Linear inequalities and the 'both' overlap

Answer 9 of 15 · Question 1 part c · 1 mark

2022 JAN · Paper 02 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

**Exact answers:**

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

**Rounding answers (where accuracy is specified in the question)**

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

## Type MS · Linear inequalities and the 'both' overlap

### Answer 9 of 15 · Question 1 part c · 1 mark

2022 JAN · Paper 02 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 1 (a) | $x > 3$ | B1<br>(1) |
| (b) | $(2x-1)(x-5)$ | M1 |
| | Critical values are $x = \frac{1}{2}$ and $x = 5$ | A1 |
| | $\frac{1}{2} < x < 5$ | B1ft<br>(3) |
| (c) | $3 < x < 5$ | B1ft<br>(1) |
| | | **Total 5 marks** |

Type MS · Linear inequalities and the 'both' overlap

Answer 10 of 15 · Question 2 part c · 1 mark

2023 JAN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

| | | |
|---|---|---|
| **2(a)** | $3x-8<5x+3 \Rightarrow x>-\frac{11}{2}$ | B1<br>[1] |
| **(b)** | $4x^2-7x+1>6-2x^2 \Rightarrow$<br>$6x^2-7x-5>0 \Rightarrow (3x-5)(2x+1)>0$<br>cv's are $x=\frac{5}{3}, \ -\frac{1}{2} \Rightarrow x<-\frac{1}{2}, \ x>\frac{5}{3}$ | M1M1<br><br>M1A1<br>[4] |
| **(c)** | $-\frac{11}{2}<x<-\frac{1}{2}, \ x>\frac{5}{3}$ | B1 ft<br>[1] |
| | **Total** | **6 marks** |

| Question | Notes | Marks |
|---|---|---|
| **2(a)** | $3x-8<5x+3$<br>$\Rightarrow x>-\frac{11}{2}$ | B1<br>[1] |
| **(b)** | For rearranging to form a 3TQ<br>$4x^2-7x+1>6-2x^2 \Rightarrow 6x^2-7x-5>0$ | M1 |
| | For method to solve their 3TQ. See general guidance for what constitutes an attempt to solve.<br>$6x^2-7x-5>0 \Rightarrow (3x-5)(2x+1)>0$ | M1 |
| | For writing down their critical values from their factorisation and the correct inequality for their cv's [outside region].<br>cv's are $x=\frac{5}{3}, \ -\frac{1}{2}$<br>$x<-\frac{1}{2}, \ x>\frac{5}{3}$ | M1 |
| | For the correct inequalities only $x<-\frac{1}{2}, \ x>\frac{5}{3}$<br>A0 if rejects one region or if inequalities are incorrectly combined. | A1<br>[4] |
| **(c)** | For the correct inequality only $-\frac{11}{2}<x<-\frac{1}{2}, \ x>\frac{5}{3}$<br>For follow through (a) must be a linear inequality and (b) must be of the form $x<a, x>b$ with $b>a$. | B1 ft<br>[1] |
| | **Total** | **6 marks** |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 10 of 15 · Question 2 part c · 1 mark

2023 JAN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

Type MS · Linear inequalities and the 'both' overlap

Answer 11 of 15 · Question 1 part c · 1 mark

2023 JAN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

**Method marks for differentiation and integration:**

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

**Use of a formula:**

Generally, the method mark is gained by **either**

Type MS · Linear inequalities and the 'both' overlap

Answer 11 of 15 · Question 1 part c · 1 mark

2023 JAN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

**or**, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

**Answers without working:**

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."**Exact answers:**

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

**Rounding answers (where accuracy is specified in the question)**

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

## Type MS · Linear inequalities and the 'both' overlap

Answer 11 of 15 · Question 1 part c · 1 mark

2023 JAN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

| **Paper 2** | | |
|---|---|---|
| Question number | Scheme | Marks |
| 1 a | $9x < 6 \Rightarrow x < \ldots$ | M1 |
| | $x < \frac{2}{3}$ | A1 |
| | | (2) |
| b | $(3x+1)(x-3) < 0$ | M1 |
| | $x = -\frac{1}{3} \quad x = 3$ | A1 |
| | $-\frac{1}{3} < x < 3$ | M1 A1 |
| | | (4) |
| c | $-\frac{1}{3} < x < \frac{2}{3}$ | B1ft |
| | | (1) |
| | | **Total 7 marks** |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 1 of 23 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 2(a) | $x < \frac{1}{2}$ o.e | B1<br>(1) |
| (b) | $10x^2 + 7x - 12 (< 0)$<br>$(5x - 4)(2x + 3) (< 0)$<br><br>$x = \frac{4}{5}$ $\quad$ $x = -\frac{3}{2}$<br><br>$-\frac{3}{2} < x < \frac{4}{5}$ o.e | M1<br><br><br><br>M1A1<br>(3) |
| (c) | $-\frac{3}{2} < x < \frac{1}{2}$ | B1FT<br>(1) |
| | | **Total 5 marks** |

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | B1 | $x < \frac{1}{2}$ o.e |
| (b) | M1 | Rearranges to a 3TQ and attempts to solve. See general guidance for what constitutes an attempt to solve.<br>If they use a calculator to solve the 3TQ, the 3TQ must be correct and the values of $x$ also correct.<br>They must obtain **two** values of $x$ for this mark |
| | M1 | For forming an **inside** region with their critical values. |
| | A1 | For the correct inequality.<br>May be seen described in set notation. |
| (c) | B1FT | For $-\frac{3}{2} < x < \frac{1}{2}$<br>FT their (a) and (b) provided they overlap and it is an inside region.<br><br>**NB** The final answer given must be consistent with their parts (a) and (b) |

| Question | Scheme | Marks |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

### Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 2 of 23 · find part c in this mark scheme extract

| number | | |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 3 of 23 · find part c in this mark scheme extract

| | |
|---|---|
| M1 | Multiplies numerator and denominator by $6+2\sqrt{5}$ with no errors |
| M1 | Equates numerator coefficients to find two simultaneous equations in $a$ and $b$ |
| M1 | Solves their simultaneous equations by any method provided the equations of the form;<br>$ma+nb=...$ and $pa+qb=...$ where $m \neq p$ and $n \neq q$<br>They must reach both a value for $a$ and a value for $b$ for this mark |
| M1 | For multiplying through by 8 or 4 to give a prime number in the denominator of $\frac{9+4\sqrt{5}}{c}$ |
| A1 | For the correct values of $a$, $b$ and $c$<br>Allow embedded values. |
| **ALT** | |
| M1 | 'Cross multiplies' through by $\left(6-2\sqrt{5}\right)$ **AND** by $c$ |
| M1 | Multiplies out both sides with no errors |
| M1 | Collects up like terms to obtain $c\left(a+b\sqrt{5}\right)=A+B\sqrt{5}$ |
| M1 | Factorises their expression to obtain $P\left(Q+R\sqrt{5}\right)$ where $P$, $Q$ and $R$ are integers<br>or deduces by inspection that $a = 7$ and $c = 2$ |
| A1 | For the correct values – accept if embedded in the expression. |

## Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 4 of 23 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 4 (a) (i) | $\frac{ar^6}{ar^2}=\frac{40}{10}$ | M1 |
| | $r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$ | M1 |
| | $r=-\sqrt{2}$ | A1 |
| (ii) | $ar^2=10\Rightarrow a=\frac{10}{'2'}$ | M1 |
| | $a=5$ | A1 (5) |
| (b) | $\lvert r\rvert>1$ so the series is not convergent. | B1FT (1) |
| | | **Total 6 marks** |

| Part | Mark | Additional Guidance |
|---|---|---|
| (a)(i) | M1 | For $\frac{ar^6}{ar^2}=\frac{40}{10}$ or $\frac{ar^2}{ar^6}=\frac{40}{10}$ |
| | M1 | For rearranging to find a value for $r$<br>$r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$ or $\left(\pm\sqrt[4]{4}\right)$ |
| | A1 | For the correct value of $r=-\sqrt{2}$ Accept $-\sqrt[4]{4}$ |
| (a)(ii) | M1 | For attempting to find a value for $a$ using their $r$<br>$ar^2=10\Rightarrow a=\frac{10}{'2'}$ |
| | A1 | For the correct value of $a=5$<br>Allow this mark from the use of $r=\sqrt{2}$ |
| (b) | B1FT | $\lvert r\rvert>1$ so the series is not convergent.<br>Follow through their value of $r$ provided it is greater than 1. |

Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 5 of 23 · find part c in this mark scheme extract

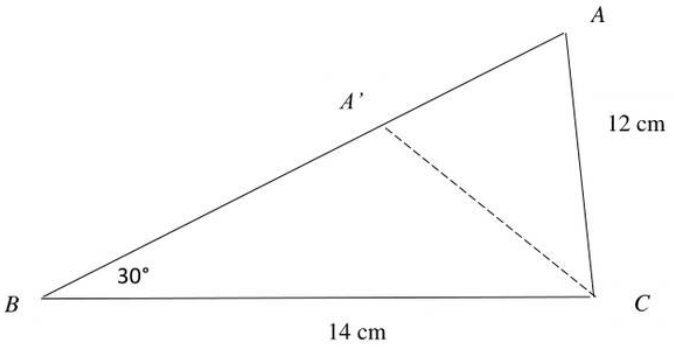
| Question number | Scheme | Marks |
|---|---|---|
| 5(a) | $12^2 = 14^2 + x^2 - 2\times 14x\cos 30$ | M1 |
| | $x^2 - 14\sqrt{3}x + 52 = 0$ | M1 A1 |
| | $x = \dfrac{14\sqrt{3} \pm \sqrt{(14\sqrt{3})^2 - 4(1)(52)}}{2(1)}$ | M1 |
| | $x = 7\sqrt{3} \pm \sqrt{95}$ | A1cso [5] |
| (b) | Area $= \dfrac{1}{2}\times 14\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \sin 30$ or uses Area $= \dfrac{1}{2}\times 12\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \dfrac{7}{12}$ | M1 |
| | Difference in areas | |
| | $D = \dfrac{1}{2}\times 14\times\left(7\sqrt{3} + \sqrt{95}\right)\times \sin 30 - \dfrac{1}{2}\times 14\times\left(7\sqrt{3} - \sqrt{95}\right)\times \sin 30$ | dM1 |
| | $D = 7\sqrt{95}$ | A1 [3] |
| **ALT Part (a) – sine rule** | | |
| | $\dfrac{12}{\sin 30} = \dfrac{14}{\sin A} \Rightarrow \sin A = \dfrac{7}{12}$ | {M1 |
| | $\left(\dfrac{7}{12}\right)^2 + \cos^2 A \equiv 1 \Rightarrow \cos A = \sqrt{1 - \left(\dfrac{7}{12}\right)^2}$ | M1 |
| | $\sin A = \dfrac{7}{12}, \cos A = \pm\dfrac{\sqrt{95}}{12}$ | A1 |
| | $x = 12\cos A + 14\cos 30 \Rightarrow x = 12\times\dfrac{\pm\sqrt{95}}{12} + 14\times\dfrac{\sqrt{3}}{2}$ | M1 |
| | $x = 7\sqrt{3} \pm \sqrt{95}$ * | A1 Cso [5]} |
| | | **Total 8 marks** |

Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 6 of 23 · find part c in this mark scheme extract

**USEFUL SKETCH**




| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | Use the cosine rule to form an equation. Rule to be correct but may be in any rearranged form. |
| | M1 | Rearrange to obtain a 3TQ. Terms in any order. = 0 may be missing. |
| | A1 | Correct 3TQ Allow poor notation – missing = 0 |
| | M1 | Attempt to solve their 3TQ by use of the formula or completing the square. See general guidance for what constitutes an attempt to solve. Use of calculators: If the 3TQ is correct, accept correct solutions without working. |
| | A1 | Correct values for $x$ $x = 7\sqrt{3} \pm \sqrt{95}$ |
| (b) | M1 | For use of area of triangle $= \frac{1}{2}ac\sin B$ with $a = 14$, $B = 30°$ and *their* $x$ May use either value found for $x$. Or uses $= \frac{1}{2}bc\sin A$ with $c = 12$ and $\sin A = \frac{7}{12}$ |
| | dM1 | For finding the difference between their two areas provided both areas are positive This M mark is dependent on the previous M mark |
| | A1 | Correct difference in areas (must be exact value) |

| | ALT Part (a) – Uses sine rule | |
|---|---|---|
| | M1 | Use of the sine rule to find sin$A$ |
| | M1 | Use of $\sin^2 A + \cos^2 A \equiv 1$ to find an exact value for $\cos A$ |
| | A1 | Correct exact values for sin$A$ and $\cos A$ |
| | M1 | Method to find $x$ $x = 12\cos A + 14\cos 30$ Allow *their* $\cos A$ |
| | A1 | Correct values for $x$ $x = 7\sqrt{3} \pm \sqrt{95}$ |

## Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 7 of 23 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 6 (a) | $700\pi = 2\pi r^2 + 2\pi rh$ | B1 |
| | $h = \dfrac{350 - r^2}{r}$ or $rh = 350 - r^2$ | M1 |
| | $V = \pi r^2 h = \pi r^2\left(\dfrac{350 - r^2}{r}\right) = \left\{\pi r\left(350 - r^2\right)\right\} = \pi r\left(350 - r^2\right)$* | M1A1* cso [4] |
| (b) | $V = 350\pi r - \pi r^3$ | |
| | $\dfrac{dV}{dr} = 350\pi - 3\pi r^2 = 0$ | M1 |
| | $\Rightarrow 350\pi = 3\pi r^2 \Rightarrow r = \sqrt{\dfrac{350}{3}} = 10.8$ (3sf) | M1A1 |
| | $\dfrac{d^2V}{dr^2} = -6\pi r$ | M1 |
| | $\dfrac{d^2V}{dr^2} < 0 \therefore$ max | A1 [5] |
| (c) | $h = \dfrac{350 - r^2}{r} = \dfrac{350 - \dfrac{350}{3}}{\sqrt{\dfrac{350}{3}}} = 21.6$ (3 s.f.) | B1 [1] |
| | | **Total 10 marks** |

Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 8 of 23 · find part c in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | B1 | For a **correct** equation for the surface area in terms of the radius and height |
| | M1 | For rearranging to find $h$ in terms of $r$ which must be as a minimum<br>$h = \frac{k - lr^2}{mr}$ where $k$, $l$ and $m$ are integers<br>OR<br>$rh = k - lr^2$ where $k$ and $l$ are integers |
| | M1 | For a correct expression for volume using *their* $h$ in terms of $r$ |
| | A1* cso | For rearranging and obtaining the given result with no errors or omissions and must include $V = \ldots$<br>[Allow Volume = …. or Vol = …..] |
| (b) | M1 | For an attempt to differentiate the **given** expression for $V$ which must be of the form $\frac{dV}{dr} = 350\pi - K\pi r^2$ where $K$ is a constant<br><br>Using Product Rule:<br><br>$\frac{dV}{dr} = \pi r(-2r) + \pi(350 - r^2) \Rightarrow \frac{dV}{dr} = \left[350\pi - K\pi r^2\right]$ |
| | M1 | For setting their $\frac{dV}{dr} = 0$ **and** attempting to find a value for $r$. |
| | A1 | $r = \sqrt{\frac{350}{3}}$ [=10.801...] accept awrt 10.8 |
| | M1 | For finding $\frac{d^2V}{dr^2}$<br>Accept only $\frac{d^2V}{dr^2} = \pm k\pi r$ where $k$ is a constant<br>See general guidance for what constitutes an attempt to differentiate. |
| | A1 | For the correct conclusion derived from the correct working throughout. |
| (c) | B1 | For obtaining the value of $h$ correct to 3 s.f.<br>Accept awrt 21.6 |

## Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 9 of 23 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 7 (a) | $x$: 0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5<br>$y$: 4.26, **4.14**, 4, **3.83**, 3.63, **3.37**, 3, 2.37 | B2<br>(2) |
| (b) | Points plotted within half a square.<br>Points joined with a smooth curve. | B1ft<br>B1ft<br>(2) |
| (c) | $3^{2x-5} = (4-x)^3 \Rightarrow 2x-5 = 3\log_3(4-x)$ | M1M1 |
| | $\frac{2x-5}{3}+3 = \log_3(4-x)+3 \Rightarrow \frac{2x+4}{3} = \log_3(4-x)+3$ | M1A1 |
| | **ALT** | |
| | $\log_3(4-x)+3 = mx+c$<br>$\Rightarrow \log_3(4-x) = mx+(c-3)$<br>$\Rightarrow (4-x) = 3^{mx+(c-3)}$ | [M1 |
| | $\Rightarrow (4-x)^3 = 3^{(mx+(c-3))3} = 3^{3mx+3c-9}$ | M1 |
| | $\Rightarrow 2x-5 = 3mx+3c-9 \Rightarrow 3m = 2$ and $-5 = 3c-9$ | M1 |
| | $\Rightarrow m = \frac{2}{3} \quad c = \frac{4}{3}$ | A1] |
| | Graph of $y = \frac{2x+4}{3}$ drawn. Intersection point is $x = 2.8$ | M1A1<br>(6)<br>**[10]** |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 10 of 23 · find part c in this mark scheme extract

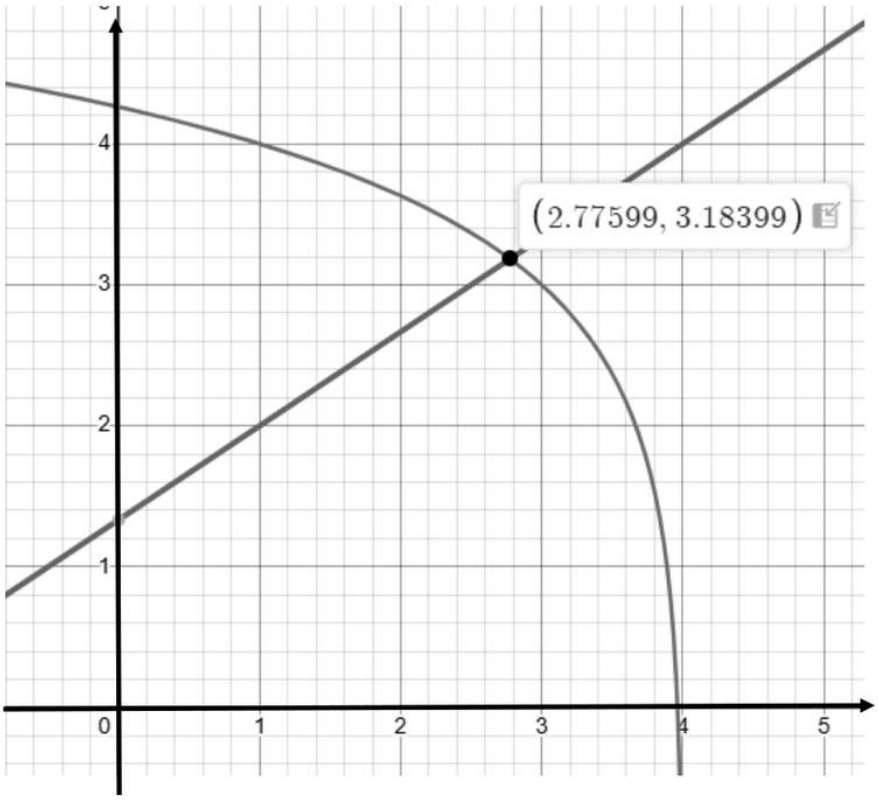
| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | B2 | All three points correct<br>(B1 for 2 points correct) |
| (b) | B1ft | Points plotted ft their table. Allow half a square tolerance. |
| | B1ft | Points joined together with a smooth curve. Allow half a square tolerance on joining up their points.<br>FT their table |
| (c) | M1 | For taking logs base 3 of both sides<br>$2x-5=\log_3(4-x)^3$ |
| | M1 | For applying the power law<br>$2x-5=3\log_3(4-x)$ |
| | ddM1 | For attempting to rearrange to the required form<br>An attempt is defined as dividing through by 3 and adding 3 to both sides. |
| | A1 | For the correct line, in any form<br>$\frac{2x-5}{3}+3=\log_3(4-x)+3\Rightarrow\left[\left(y=\right)\frac{2}{3}x+\frac{4}{3}\right]$ |
| | **ALT for first 4 marks** | |
| | M1 | Sets up the equation to obtain $\log_3(4-x)+3=mx+c$<br>And removes the log to obtain $(4-x)=3^{mx+(c-3)}$ |
| | M1 | Raises **both** sides to a power of 3.<br>Allow one error in this on the LHS |
| | M1 | Compares coefficients to obtain an equation in $m$ and an equation in $c$ |
| | A1 | Obtains the values $m=\frac{2}{3}$ $c=\frac{4}{3}$ |
| | **Draws the line** | |
| | M1 | For their line drawn provided it is of the form $y=kx\pm\frac{4}{3}$ for $k\neq 0,\ 1$ |
| | A1 | $x=2.8$ or 2.7<br>Calculator value is 2.776 |

## Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 11 of 23 · find part c in this mark scheme extract

**USEFUL GRAPH**



Type MS · Linear inequalities and the 'both' overlap

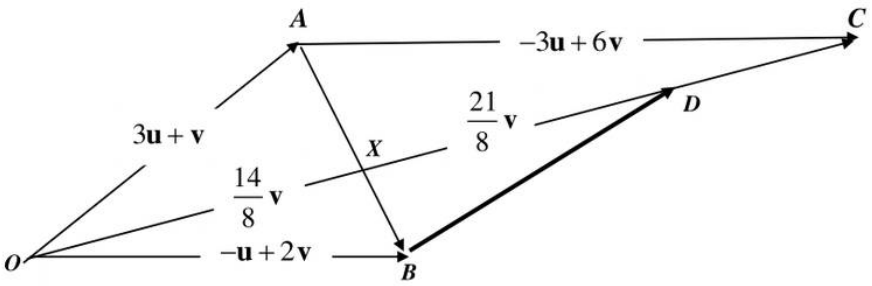
Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 12 of 23 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 8 (a) | $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}, = -(3\mathbf{u}+\mathbf{v}) + (2\mathbf{v}-\mathbf{u}) = -4\mathbf{u}+\mathbf{v}$ | M1,A1<br>(2) |
| (b) | $\overrightarrow{AC} = \lambda(2\mathbf{v}-\mathbf{u})$ | M1 |
| | $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = (3\mathbf{u}+\mathbf{v}) + \lambda(2\mathbf{v}-\mathbf{u}) = \mu\mathbf{v}$ | M1 |
| | $\mathbf{u}: 0 = 3-\lambda \qquad \mathbf{v}: \mu = 1+2\lambda$ | M1 |
| | $\mu = 7 \qquad [\lambda = 3]$ | A1<br>(4) |
| (c) | $\overrightarrow{OX} = \overrightarrow{OA} + \alpha\overrightarrow{AB} = 3\mathbf{u}+\mathbf{v}+\alpha(-4\mathbf{u}+\mathbf{v})$<br>or<br>$\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AC} + \beta\overrightarrow{CO} = 3\mathbf{u}+\mathbf{v}+3(2\mathbf{v}-\mathbf{u})+\beta(-7\mathbf{v})$ | M1 |
| | $\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$ | A1 |
| | $\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$ | M1 |
| | $\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$ | M1 |
| | $\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{35}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{7}{4}\mathbf{v}+\frac{21}{8}\mathbf{v}$<br>$\overrightarrow{BD} = \mathbf{u}+\frac{19}{8}\mathbf{v}$ | A1<br>(5) |
| | | **Total 11 marks** |

**USEFUL SKETCH**

$$\overrightarrow{OD} = \frac{35}{8}\mathbf{v} \quad \text{and} \quad \overrightarrow{AX} = -3\mathbf{u}+\frac{3}{4}\mathbf{v}$$



Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 13 of 23 · find part c in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | Use of $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$ |
| | A1 | $-4\mathbf{u} + \mathbf{v}$ |
| (b) | M1 | For use of $\overrightarrow{AC}$ is parallel to $\overrightarrow{OB}$ to form an expression for $\overrightarrow{AC}$ |
| | M1 | For use of their expression for $\overrightarrow{AC}$ to form an expression for $\overrightarrow{OC}$ |
| | M1 | For equating for **u** and for **v** |
| | A1 | For $\mu = 7$ |
| (c) | | **General principles for marking part (c)**<br>For example: You may see the vector path for $\overrightarrow{BD}$ first - score the 4th M mark<br>• Candidates will need to find vector $\overrightarrow{OX}$ [M1A1]<br>• Candidates will need to use the ratio of bases to find<br>$\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$ **leading** to establish<br>$\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$ [M1]<br>• Check any path they choose on the diagram below. For example,<br>$\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OX} + \overrightarrow{XD}$<br>or [M1]<br>$\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OA} + \overrightarrow{AX} + \overrightarrow{XD}$<br>If a candidate chooses to take a longer path and need to find for example, $\overrightarrow{AX}$ they will need to find a vector. There is no separate mark for this.<br>• For the correct vector |
| | M1 | For an expression for $\overrightarrow{OX}$ including an unknown constant |
| | A1 | For $\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$ |
| | M1 | For recognising that triangle $OBX$ and triangle $OBD$ share a base $OB$ and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{OD}$ together with an attempt to find $\overrightarrow{OD}$<br>OR<br>For recognising that triangle $OBX$ and triangle $BXD$ share a base $BX$ and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{XD}$ together with an attempt to find $\overrightarrow{XD}$ |
| | M1 | For method to find $\overrightarrow{BD}$ Award this for any valid path given |
| | A1 | $\overrightarrow{BD} = \mathbf{u} + \frac{19}{8}\mathbf{v}$ |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 14 of 23 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 9 (a) | $\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ * | M1A1A1 Cso [3] |
| (b) | **Method A** | |
| | $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ | M1 |
| | $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$ | M1 |
| | $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$ | M1 |
| | $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y \Rightarrow \frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$<br>$\Rightarrow \frac{d^2y}{dt^2} = -8\frac{dy}{dt} - 20y \Rightarrow \frac{d^2y}{dt^2} + 8\frac{dy}{dt} + 20y = 0$ | M1 |
| | $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = 0 \Rightarrow M = 8, N = 20$ | A1 [5] |
| | **Method B** | |
| | $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$<br>$\left(= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t\right)$ | [M1 |
| | $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$ | |
| | $= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$ | M1 |
| | $= (16 - 2M)e^{-4t}\sin 2t + (12 - 4M + N)e^{-4t}\cos 2t$ | M1 |
| | $16 - 2M = 0$ and $12 - 4M + N = 0$ | M1 |
| | $M = 8,\ N = 20$ | A1] |
| | **Method C**<br>By inspection – please see additional guidance. | |
| | | **Total 8 marks** |

| Part | Mark | Additional Guidance |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

### Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 15 of 23 · find part c in this mark scheme extract

| | | |
|---|---|---|
| (a) | M1 | For an attempt to differentiate the given expression.<br>• There needs to be an acceptable attempt to differentiate both terms. $e^{-4t}\cos 2t \rightarrow ke^{-4t}\sin 2t + le^{-4t}\cos 2t$ with $k, l \neq 0$<br>• There need to be two terms added.<br>$\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t$ |
| | A1 | One term completely correct.<br>$\frac{dy}{dt} = -4e^{-4t}\cos 2t + '-2e^{-4t}\sin 2t$ OR $\frac{dy}{dt} = '-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t$ |
| | A1 | Fully correct differentiated expression and rearranged into the required form<br>$\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$<br>Accept also,<br>For example:<br>$-\left(-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t\right) - 4\left(e^{-4t}\sin 2t\right) = 2e^{-4t}\sin 2t$<br>With any unambiguous indication that a conclusion has been reached, e.g., QED, hash sign, double underline, RHS = LHS etc. |
| | | **Method A** |
| (b) | M1 | For an attempt to find $\frac{d^2y}{dt^2}$<br>Minimally acceptable attempt is<br>$ke^{-4x}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$<br>$k, l$ as in their first derivative, $m, n, p, q \neq 0$<br>$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ |
| | M1 | For substituting $y$ and $\frac{dy}{dt}$ into their $\frac{d^2y}{dt^2}$<br>$\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$ |
| | M1 | For preparing to eliminate $\cos 2t$ by rearranging their $\frac{dy}{dt}$<br>$2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$<br>Allow errors in arithmetic but not mathematically incorrect process. |
| | M1 | For an unsimplified expression only in terms of $y$, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$<br>$\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$ |
| | A1<br>[8] | For finding the value of $M$ and the value of $N$.<br>$M = 8,\ N = 20$ |
| | | **Method B** |

## Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

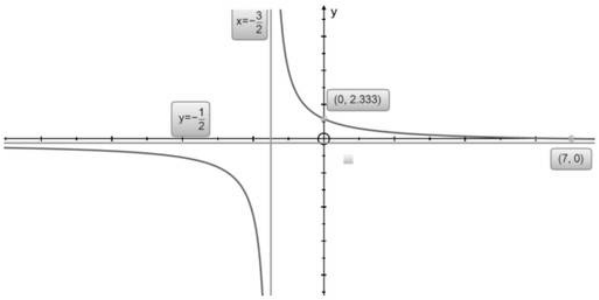
2025 JUN · Paper 02 · extract 16 of 23 · find part c in this mark scheme extract

| | | |
|---|---|---|
| | M1 | For an attempt to find $\frac{d^2y}{dt^2}$<br>Minimally acceptable attempt is<br>$ke^{-4t}\sin 2t + le^{-4t}\cos 2t \to k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$<br>$k, l$ as in their first derivative, $m, n, p, q \neq 0$<br>$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ |
| | M1 | For substituting $y$, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ into the equation.<br>$\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$<br>$= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$ |
| | M1 | For collecting terms<br>$\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = ('16' - 2M)e^{-4t}\sin 2t + ('12' - 4M + N)e^{-4t}\cos 2t$ |
| | M1 | For identifying two equations from their working and a method to solve them simultaneously<br>For example, $'16' - 2M = 0$ and $'12' - 4M + N = 0$ |
| | A1 | For finding the value of $M$ and the value of $N$.<br>$M = 8, N = 20$ |
| | | **Method C (By inspection)** |
| | M1 | For an attempt to find $\frac{d^2y}{dt^2}$<br>Minimally acceptable attempt is<br>$ke^{-4t}\sin 2t + le^{-4t}\cos 2t \to k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$<br>$k, l$ as in their first derivative, $m, n, p, q \neq 0$<br>$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ |
| | M1 | For simplifying $\frac{d^2y}{dt^2} = e^{-4t}16\sin 2t + e^{-4t}12\cos 2t = \left[e^{-4t}(16\sin 2t + 12\cos 2t)\right]$ |
| | M1 | For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}$ $[\sin 2t]$ and deducing that $'16' + (\underline{8} \times -2) = 0 \Rightarrow M = 8$ |
| | M1 | For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}$ and $y$ and deducing that $'12' - ('8' \times 4) + \underline{20} = 0 \Rightarrow N = 20$ |
| | A1 | For both $M = 8$ and $N = 20$ |
| | | **Total 8 marks** |

Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 17 of 23 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 10 (a) | (i) $x=-\frac{3}{2}$ (ii) $y=-\frac{1}{2}$ | B1<br>B1<br>[2] |
| (b) | $(7,0)$<br>$\left(0,\frac{7}{3}\right)$ | B1<br>B1<br>[2] |
| (c) |  | B1ft curve<br>B1ft asymptotes<br>B1ft (intersections with $x$- and $y$-axes)<br>[3] |
| (d) | $\frac{-1(2x+3)-2(7-x)}{(2x+3)^2}=\left[\frac{-17}{(2x+3)^2}\right]$ | M1 |
| | $\frac{-1\left(2\times\frac{1}{2}+3\right)-2\left(7-\times\frac{1}{2}\right)}{\left(2\times\frac{1}{2}+3\right)^2}=-\frac{17}{16}$ | M1A1* cso<br>[3] |
| | ALT – product rule | |
| | $-1(2x+3)^{-1}+(7-x)(-1)(2)(2x+3)^{-2}=\left[\frac{-17}{(2x+3)^2}\right]$ | M1 |
| | $-1\left(2\times\frac{1}{2}+3\right)^{-1}+\left(7-\frac{1}{2}\right)(-1)(2)\left(2\times\frac{1}{2}+3\right)^{-2}=-\frac{17}{16}$ | M1A1* cso |

## Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 18 of 23 · find part c in this mark scheme extract

| | | |
|---|---|---|
| (e) | $-\frac{17}{(2x+3)^2}=-\frac{17}{16}$ | M1 |
| | $(2x+3)^2=16\Rightarrow 2x+3=\pm\sqrt{16}\Rightarrow x=0.5 \text{ or } -3.5$ | dM1 |
| | $y=\frac{7--3.5}{2\times-3.5+3}\left(=-\frac{21}{8}\right)$ | ddM1 |
| | $\left(-\frac{7}{2},-\frac{21}{8}\right)$ | A1 |
| | $y--\frac{21}{8}=-\frac{17}{16}\left(x--\frac{7}{2}\right)$ | M1A1 |
| | $34x+32y+203=0$ | A1<br>[7] |
| | | **Total 17 marks** |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 19 of 23 · find part c in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| **For parts (a) and (b) look for their responses in the body of the question. In part (a), if the responses are not labelled (i) and (ii) accept them in order.** | | |
| (a) | (i) B1 | For $x=-\frac{3}{2}$ oe |
| | (ii) B1 | For $y=-\frac{1}{2}$ oe |
| (b) | B1 B1 | First B1 for either correct, second B1 for both correct<br>Condone if not given as coordinates e.g. $x=7$ and/or $y=\frac{7}{3}$ given |
| (c) | B1ft | Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes. |
| | B1ft | Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes. |
| | B1ft | Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around.<br>If additional intersections seen then B0 |
| (d) | M1 | Attempt the quotient rule. Numerator must be of the form $A.(2x+3)-B.(7-x)$, $B>1$.<br>Denominator must be of the form $(2x+3)^2$ |
| | M1 | Substitutes $x=\frac{1}{2}$ into their derivative<br>**NB: This is an A mark in Epen** |
| | A1* cso | Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors<br>**NB: This is a B mark in Epen** |
| | ALT – product rule | |
| | M1 | For an attempt at Product Rule.<br>Must be a sum of two products.<br>Must have the form<br>$c(2x+3)^{-1}+d(7-x)(2x+3)^{-2}$ for constants $c$, $d$. |
| | M1 | Substitutes $x=\frac{1}{2}$ into their derivative<br>**NB: This is an A mark in Epen** |
| | A1* cso | Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors<br>**NB: This is a B mark in Epen** |
| (e) | M1 | For equating the gradient of the tangent to their $\frac{dy}{dx}$ from (d) |
| | dM1 | For rearranging to obtain $(2x+3)^2=16$ and attempt to solve<br>or for rearrangement to a 3TQ and method to solve the 3TQ. See general guidance for acceptable methods to solve. |

## Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 20 of 23 · find part c in this mark scheme extract

| | | |
|---|---|---|
| | ddM1 | For substitution of their $x$ into the original equation to obtain $y$.<br>This must **not** be $x=\frac{1}{2}$<br>This M mark is dependent on the **two** previous M marks |
| | A1 | For correct coordinate of $P\left(-\frac{7}{2},-\frac{21}{8}\right)$ |
| | M1 | For substitution of their coordinates of $P$ and the gradient $-\frac{17}{16}$ to obtain an equation of a straight line using any valid method.<br>If $y=mx+c$ is used a value of $c$ must be found **and** an equation needs to be formed.<br>For example; $'-\frac{21}{8}'=-\frac{17}{16}\times'-\frac{7}{2}'+c\Rightarrow c=-\frac{203}{32}\Rightarrow y=-\frac{17}{16}x-\frac{203}{32}$ [ |
| | A1 | For any correct equation of the tangent with the correct values of $x$ and $y$ in any form. |
| | A1 | For the correct equation of tangent given in the required form.<br>$34x+32y+203=0$ |

Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 21 of 23 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 11 (a) | $\cos^4\theta-\sin^4\theta=(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$ | M1 |
| | $=(\cos^2\theta-\sin^2\theta)\times1=\cos2\theta$ * | M1M1A1 cso |
| | **ALT** example | [4 |
| | $\cos^4\theta-\sin^4\theta=\cos^4\theta-\sin^2\theta\sin^2\theta$ | |
| | $=\cos^4\theta-(1-\cos^2\theta)(1-\cos^2\theta)$ | 2nd M1 |
| | $=\cos^4\theta-(1-2\cos^2\theta+\cos^4\theta)$ | 1st M1 |
| | $=(1-2\cos^2\theta)$ | 3rd M1 |
| | $=\cos2\theta$ | A1 |
| (b) | $8\cos^2\left(2\theta+\frac{\pi}{4}\right)-3=2[\cos^4\left(\theta+\frac{\pi}{8}\right)-\sin^4\left(\theta+\frac{\pi}{8}\right)]$ | M1 |
| | $=2\cos\left(2\theta+\frac{\pi}{4}\right)$ | A1 |
| | $8\cos^2\left(2\theta+\frac{\pi}{4}\right)-2\cos\left(2\theta+\frac{\pi}{4}\right)-3=0$ | M1 |
| | $\left(4\cos\left(2\theta+\frac{\pi}{4}\right)-3\right)\left(2\cos\left(2\theta+\frac{\pi}{4}\right)+1\right)[=0]$ | |
| | $\cos\left(2\theta+\frac{\pi}{4}\right)=\frac{3}{4}\quad\cos\left(2\theta+\frac{\pi}{4}\right)=-\frac{1}{2}$ | A1 |
| | $2\theta+\frac{\pi}{4}=0.72273...\quad2\theta+\frac{\pi}{4}=\frac{2\pi}{3}$ | dM1 |
| | $\theta=2.39,3.11,\ 0.65,1.70$ accept $\theta=\frac{5\pi}{24},\frac{13\pi}{24}$ | A1A1 [7] |
| (c) | $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos^4 2x-\sin^4 2x-8\sin 4x\right)\mathrm{d}x=\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos 4x-8\sin 4x\right)\mathrm{d}x$ | M1 |
| | $=\left[\frac{1}{4}\sin 4x+\frac{8}{4}\cos 4x\right]_{\frac{\pi}{16}}^{\frac{\pi}{8}}$ | M1 |
| | $=\frac{1}{4}\left[\sin\left(4\times\frac{\pi}{8}\right)-\sin\left(4\times\frac{\pi}{16}\right)\right]+2\left[\cos\left(4\times\frac{\pi}{8}\right)-\cos\left(4\times\frac{\pi}{16}\right)\right]$ | M1 |
| | $=\frac{1}{4}\left[1-\frac{\sqrt{2}}{2}\right]+2\left[0-\frac{\sqrt{2}}{2}\right]=\frac{1}{4}-\frac{9\sqrt{2}}{8}$ | A1 [4] |
| | | **Total 15 marks** |

Type MS · Linear inequalities and the 'both' overlap

Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 22 of 23 · find part c in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| **General Principles for marking part (a)**<br>• First M mark is for either applying the difference of two squares or multiplying out their expression and collecting terms.<br>• Second M mark is for applying $\cos^2\theta+\sin^2\theta=1$ in any form. This can be implied from correct work. However, do not award for just $\cos^4\theta-\sin^4\theta=\cos^2\theta-\sin^2\theta$ seen.<br>• Third M mark is for applying any identity for $\cos 2\theta$<br>• A1 For obtaining the required result with no errors. | | |
| (a) | M1 | For factorising as the difference of two squares<br>$(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$ |
| | M1 | For use of $\cos^2\theta+\sin^2\theta)=1$ |
| | M1 | For use of $\cos^2\theta-\sin^2\theta=\cos(2\theta)$ |
| | A1 cso | Obtain the given result with no errors in the working. |
| (b) | M1 | For using the given result to obtain a quadratic in $\left(2\theta+\frac{\pi}{4}\right)$<br>Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark |
| | A1 | For rearranging to the correct quadratic |
| | M1 | For a valid attempt to solve their **3TQ**.<br>See General Guidance for what constitutes an attempt to factorise.<br>If candidates use a calculator [so no working shown] then award this mark only if the 3TQ is correct with the correct solutions. |
| | A1 | For obtaining correct values for $\cos\left(2\theta+\frac{\pi}{4}\right)$<br>Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark |
| | dM1 | For achieving any valid value **in radians** for their $2\theta+\frac{\pi}{4}$<br>i.e. 0.7227…or $\frac{2}{3}\pi$<br>Accept values out of range for this mark.<br>This is dependent on the previous M mark. |
| | A1 | For two correct solutions in range |
| | A1 | For all four correct solutions in range and no others.<br>Extra solutions within range scores A0<br>Ignore solutions outside of range.<br>Exact solutions when given must be given as multiples of π. |
| (c) | M1 | For use of the given result to obtain $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\cos 4x\ \mathrm{d}x$ |
| | M1 | For an attempt to integrate<br>$\cos kx\to l\sin kx$ where $k,l\neq 1$ |
| | M1 | For substituting limits into a changed expression |
| | A1 | For the correct answer in the required form. |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 12 of 15 · Question 2 part c · 1 mark

2025 JUN · Paper 02 · extract 23 of 23 · find part c in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

## Type MS · Linear inequalities and the 'both' overlap

### Answer 13 of 15 · Question 1 part c · 1 mark

2021 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

| Number | | |
|---|---|---|
| 1 (a) | $3x < 12$ | M1 |
| | $x < 4$ | A1 |
| | | [2] |
| (b) | $(2x+1)(x-3) > 0$ | M1 |
| | Critical values are $x = -\frac{1}{2}$ and $x = 3$ | M1 |
| | $x < -\frac{1}{2}$ $x > 3$ | A1 |
| | | [3] |
| (c) | $x < -\frac{1}{2}$ $3 < x < 4$ | B1ft |
| | | [1] |
| | | **Total 6 marks** |

## Type MS · Linear inequalities and the 'both' overlap

Answer 13 of 15 · Question 1 part c · 1 mark

2021 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

| Part | Mark | Guidance |
|---|---|---|
| (a) | M1 | Attempts to solve the inequality to achieve $3x < 12$<br>Allow $3x < a$ where $a$ is an integer |
| | A1 | For $x < 4$ |
| (b) | M1 | Attempts to solve the inequality by any method to find critical values<br>See General Guidance for acceptable methods.<br>If a calculator is used, the critical values must be fully correct for this mark.<br><br>Allow = or > for this mark or even no sign at all provided it is clear they are solving a quadratic.<br>$(2x+1)(x-3) > 0 \Rightarrow x = ...,\ ...$ $\left(x = -\frac{1}{2},\ 3\right)$ |
| | M1 | For forming a correct inequality, which must be an open interval, following through their **two** critical values which must have come from the solution of a 3TQ.<br>$x < -\frac{1}{2} \quad x > 3$<br>Accept any correct notation. E.g., $x < -\frac{1}{2}$ or $x > 3$<br>Or $\left\{x : x < -\frac{1}{2}\right\} \cup \{x : x > 3\}$<br>Condone $x < -\frac{1}{2}$ and $x > 3$ for this mark only |
| | A1 | For the correct inequality with the correct critical values using any acceptable notation.<br>Eg, $x < -\frac{1}{2} \quad x > 3$ |
| (c) | B1ft | For $x < -\frac{1}{2} \quad 3 < x < 4$<br>ft their answers from parts (a) and (b)<br>provided (b) is of the form $x < p$ and/or $x > q$<br><br>**Note:** If you have already penalised them for writing $x < -\frac{1}{2}$ **and** $x > 3$ in part (b) then allow $x < -\frac{1}{2}$ **and** $3 < x < 4$ for this mark. |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 13 of 15 · Question 1 part c · 1 mark

2021 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

## Type MS · Linear inequalities and the 'both' overlap

Answer 14 of 15 · Question 2 part c · 1 mark

2023 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

| | | |
|---|---|---|
| **2(a)** | $2(x+1)<5x-2 \Rightarrow 2x+2<5x-2 \Rightarrow 3x>4 \Rightarrow x>\frac{4}{3}$ | M1A1<br>[2] |
| **(b)** | $3x^2-x-10=0 \Rightarrow (3x+5)(x-2)=0 \Rightarrow x=-\frac{5}{3}, 2$<br>$'-\frac{5}{3}' \leqslant x \leqslant '2'$ allow $'-\frac{5}{3}' < x < '2'$<br>$-\frac{5}{3} \leqslant x \leqslant 2$ | M1<br>dM1<br>A1<br>[3] |
| **(c)** | $"\frac{4}{3}" < x \text{ '}\leqslant\text{'} "2"$ | B1ft<br>[1] |
| | | **Total 6 marks** |

| Part | Mark | Notes |
|---|---|---|
| (a) | M1 | Attempts to solve the inequality with no more than one arithmetical error. |
| | A1 | For the correct inequality. |
| (b) | M1 | For attempting to solve the QE to find **two** critical values. |
| | For the dM and A marks allow any acceptable notation.<br>For example; $-\frac{5}{3} \leqslant x \cap x \leqslant 2$ or, $-\frac{5}{3} \leqslant x$ **and** $x \leqslant 2$<br>The region must however indicate an INSIDE region | |
| | dM1 | Forms an inside region with **their** cv's<br>This mark is dependent on the first M mark in (b) |
| | A1 | For the correct region with the correct inequalities. |
| (c) | B1ft | For the correct combined inside region.<br>Penalise the incorrect inequality from part (b) $<$ in place of $\leqslant$ only once.<br>Even if the **correct** inequality does not follow from their work, award this mark. |

## Type MS · Linear inequalities and the 'both' overlap

### Answer 14 of 15 · Question 2 part c · 1 mark

2023 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

Type MS · Linear inequalities and the 'both' overlap

Answer 15 of 15 · Question 2 part c · 1 mark

2020 JAN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(a) | $x \leqslant -1$ | B1 | (1) |
| (b) | $8x^2 + 10x - 3 (< 0)$ | | |
| | $(4x - 1)(2x + 3) (< 0)$ | M1 | |
| | $x = \frac{1}{4} \quad x = -\frac{3}{2}$ | A1A1 | |
| | $-\frac{3}{2} < x < \frac{1}{4}$ | A1ft | (4) |
| (c) | $-\frac{3}{2} < x \leqslant -1$ | B1 | (1) |
| | | | **[6]** |

(a)
B1 For $x \leqslant -1$

(b) Accept decimals in (b) and (c)
**NB** The first 3 marks are for finding the critical values. Allow with < or = used.
M1 Attempt to obtain the critical values by solving their 3TQ by any valid method.
A1 Either CV correct
A1 Second CV correct. Award these 2 marks if correct CVs seen in an inequality.
A1ft Inequality formed to indicate the values between their CVs. Must use < (Can be written in set language).
**NB** If CVs incorrect and only shown in the inequality, award 0/4 if no working shown for solving their 3TQ: if working shown M1A0A0A1 is available.

(c)
B1 For $-\frac{3}{2} < x \leqslant -1$ (no ft)

## Type MS · Linear inequalities and the 'both' overlap

### Answer 15 of 15 · Question 2 part c · 1 mark

2020 JAN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

| Question Number | Scheme | Marks |
|---|---|---|